

By Friday, March 11, 11:59pm, you need to submit your brief plan for the term project. The plan should include your project option (development or research), your team members (up to four students), and the topic (brief description of what you plan to do for the project, in one or two paragraphs).

Main Overview And Main Goal:

Our envisioned project is an encrypted file storage system that is built on Amazon S3 with access control rules. Users will be able to upload and download files through our application, and the system will securely store these files in S3. Before a file is uploaded to the system, it will be encrypted with an encryption algorithm so that its contents are protected when it is in the cloud. Our goal with this project is to create a secure and logical cloud storage system that can implement confidentiality and ensure that authorized users only have access to protected files that they were meant to access. Users may have their own files, and some may be shared with selected users, yet the system will check permissions before allowing downloads or viewing.

How Security will work and how it will be implemented:

- Client-side or application side encryption
- Secret key handling
- User authentication
- Access control rules
- S3

Security in our system will work across a few layers using what we have covered in class and labs. Files will be encrypted on the client side before being uploaded to S3, so the storage layer never holds plaintext data. Each user will have their own encryption key, and shared files will only be accessible to explicitly authorized users. We will use AWS Cognito or IAM for user authentication, so only verified users can upload or download files, and S3 bucket policies will make sure users can only access files they own or have been given permission to view. This project is essentially us taking the S3, IAM, and encryption concepts from the course and building them into one working secure storage system.